

HERMISTON, OR, USA

WMO: 726883

Lat: 45.826N

Lon: 119.261W

Elev: 641

StdP: 14.36

Time Zone: -8.00 (NAP)

Period: 98-19

WBAN: 04113

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	10.2	16.9	3.1	6.6	13.7	9.7	9.2	19.3	27.5	49.7	24.7	48.4	3.4	160	0.560

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.2	99.1	67.0	95.8	65.6	92.0	64.5	69.0	93.8	67.4	91.0	65.9	88.6	6.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	81.2	75.6	58.2	74.0	74.7	56.5	69.7	74.3	33.4	93.9	32.0	90.7	30.8	88.4	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.7	18.3	3.4	104.8	9.5	3.0	-3.5	106.9	-9.0	108.7	-14.3	110.3	-21.2	112.5
			2.6	72.7	9.4	4.0	-4.1	75.5	-9.6	77.8	-14.8	80.0	-21.6	82.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.3	35.2	38.2	45.7	51.9	60.4	67.5	75.4	73.6	64.2	52.0	40.9	34.1	(6)
	DBStd	15.86	8.56	7.51	6.54	5.52	6.92	6.22	5.84	5.45	6.00	6.89	8.92	9.18	(7)
	HDD50	1848	461	333	162	41	4	0	0	0	1	56	291	499	(8)
	HDD65	5061	924	749	599	393	178	41	2	5	85	404	722	959	(9)
	CDD50	3071	2	4	28	99	326	524	788	731	426	119	19	5	(10)
	CDD65	808	0	0	0	1	35	114	325	271	61	1	0	0	(11)
	CDH74	12142	0	0	1	65	723	1811	4662	3738	1096	45	1	0	(12)
	CDH80	6041	0	0	0	7	266	796	2602	1953	411	6	0	0	(13)
(14) Wind	WSAvg	7.1	5.4	6.3	8.0	8.7	8.4	9.1	8.1	7.5	6.5	6.1	5.9	5.3	(14)
(15) Precipitation	PrecAvg	9.2	1.5	0.9	0.9	0.8	0.8	0.5	0.2	0.4	0.5	0.6	1.5	1.3	(15)
	PrecMax	14.7	9.6	3.7	2.8	2.2	3.3	1.6	0.9	1.8	2.8	1.7	6.9	3.5	(16)
	PrecMin	4.4	0.2	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	(17)
	PrecStd	2.1	1.5	0.7	0.6	0.6	0.7	0.4	0.2	0.5	0.6	0.4	1.2	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.4	61.0	71.8	80.7	92.9	99.3	104.6	101.7	93.4	79.4	67.8	62.3	(19)
		MCWB	49.3	50.3	54.5	59.6	63.9	67.6	69.2	67.1	66.2	59.2	55.3	51.3	(20)
	2%	DB	54.2	56.8	65.7	74.9	87.3	93.4	100.0	97.9	88.8	73.0	62.7	55.6	(21)
		MCWB	46.6	47.8	51.7	56.8	62.4	65.4	67.2	66.6	64.1	57.5	52.5	48.6	(22)
	5%	DB	50.4	53.9	62.4	70.2	81.8	88.6	97.2	94.1	84.6	69.2	57.5	50.7	(23)
		MCWB	44.5	45.9	49.8	54.3	59.9	63.1	65.8	65.3	62.0	55.4	49.4	45.1	(24)
	10%	DB	46.8	50.2	58.7	65.6	77.1	84.0	93.2	90.7	80.9	65.1	54.1	46.3	(25)
		MCWB	42.2	43.2	47.9	51.6	58.0	61.3	64.7	64.2	60.4	53.4	47.5	42.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.9	51.7	56.0	60.9	65.9	70.1	71.4	71.4	68.3	62.9	59.0	52.6	(27)
		MCDB	56.5	58.9	66.2	76.3	88.8	94.3	98.7	94.3	89.2	75.1	66.2	60.6	(28)
	2%	WB	47.3	48.7	53.1	57.8	63.4	66.9	69.1	68.9	65.6	59.3	53.3	49.0	(29)
		MCDB	52.8	55.5	63.2	72.3	82.8	90.3	95.3	92.5	85.4	70.4	59.8	55.5	(30)
	5%	WB	45.2	46.2	50.9	55.2	61.4	64.5	67.5	67.1	63.4	56.8	50.7	45.7	(31)
		MCDB	49.8	52.6	60.2	68.5	78.9	85.5	92.5	89.7	81.8	66.8	56.4	50.0	(32)
	10%	WB	42.6	44.2	48.8	52.8	59.3	62.4	65.9	65.4	61.4	54.6	48.0	42.2	(33)
		MCDB	46.9	49.5	57.7	63.9	73.7	81.0	89.8	87.3	78.6	63.5	53.3	46.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	18.9	23.3	25.8	28.0	28.2	32.2	32.0	30.7	24.9	17.6	12.8	(35)
		MCDBR	18.2	21.6	27.9	34.0	36.7	37.2	38.3	37.9	37.5	29.5	21.1	18.1	(36)
	5% WB	MCWBR	12.2	13.9	15.8	17.4	15.0	14.1	13.7	14.6	16.7	16.1	12.9	11.9	(37)
		MCWBR	17.3	20.1	26.0	31.7	32.1	34.7	35.2	34.7	33.2	26.4	19.5	17.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.292	0.294	0.308	0.341	0.357	0.358	0.358	0.358	0.342	0.315	0.298	0.288	(40)
		taud	2.552	2.563	2.536	2.417	2.393	2.415	2.419	2.415	2.443	2.526	2.540	2.542	(41)
	Ebn at Noon		263	285	295	291	289	288	286	282	278	272	257	252	(42)
		Edh at Noon	20	24	28	35	36	36	36	34	30	24	20	18	(43)
(44) All-Sky Solar Radiation	RadAvg		397	714	1119	1649	1998	2277	2444	2060	1525	942	499	322	(44)
	RadStd		44	88	86	104	145	115	82	114	92	70	42	37	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	N/A	(46)
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	N/A	(47)

Nomenclature: See separate page